

1 Quick Review

This week you covered the tail end of the basis of linear algebra and began working with determinants.

There are four special types of matrices:

Definition 1: Diagonal Matrix

The lower triangular matrix is a matrix where every element of the form $a_{ij} = 0 \forall \{i, j \leq n \mid i \neq j\}$.

$$\mathbf{D} = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}$$

Definition 2: Lower & Upper Triangular Matrix

The lower triangular matrix is a matrix where everything *above* the diagonal must be 0.

The upper triangular matrix is a matrix where everything *below* the diagonal must be 0.

$$\mathbf{L} = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & u_{nn} \end{bmatrix}$$

Definition 3: Symmetric Matrix

A matrix is symmetric when $\mathbf{A} = \mathbf{A}^T$. For example, the following matrix is *symmetric*:

$$\mathbf{D} = \begin{bmatrix} 2 & 0 & 5 & 0 \\ 0 & 6 & 13 & 9 \\ 5 & 13 & 4 & 1 \\ 0 & 9 & 1 & 20 \end{bmatrix}$$

In other words, $a_{ij} = a_{ji}$.

Lastly, linear (or matrix) transformations will come up again and again throughout this course. The two key properties of a linear transformation you will want to remember include:

- **Homogeneity:** $T(k\mathbf{u}) = kT(\mathbf{u})$
- **Additivity:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$

If you can successfully prove that a transformation follows these, then they must be a linear (or matrix) transformation.

Notationally, we represent a transformation T that maps an input from $\mathbb{R}^m$ to an output in $\mathbb{R}^n$ as

$$T: \mathbb{R}^m \rightarrow \mathbb{R}^n$$

If $n = m$, then the transformation is called an operator. Familiar operators are d/dt or $\int$. An operator *does not* have to be linear.

2 Problems

1. Use the following systems to (i) Find $\mathbf{A}$, $\vec{x}$, and $\vec{b}$ and (ii) Solve the system using $\mathbf{A}^{-1}$.

(a)

$$\begin{cases} 5x_1 + 6x_2 = 33 \\ 8x_1 + 9x_2 = 34 \end{cases}$$

$$\mathbf{A} = \begin{bmatrix} & \end{bmatrix} \quad \vec{x} = \begin{bmatrix} \\ \end{bmatrix} \quad \vec{b} = \begin{bmatrix} \\ \end{bmatrix}$$

(b)

$$\begin{cases} c_1t_1 & & & = b_1 \\ & c_2t_2 & & = b_2 \\ & & c_3t_3 & = b_3 \\ & & & c_4t_4 = b_4 \end{cases}$$

$$\mathbf{A} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad \vec{x} = \begin{bmatrix} \\ \\ \\ \end{bmatrix} \quad \vec{b} = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

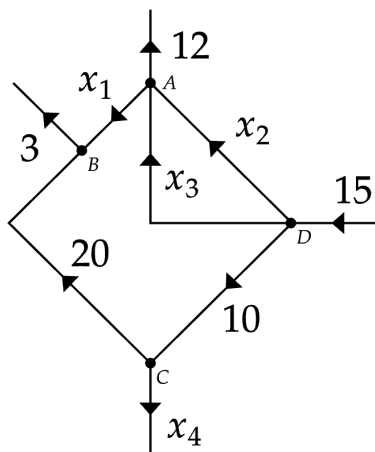
2. Below are a set of conceptual questions.

(a) “ $\mathbf{A}$ is an $m \times n$ diagonal matrix.” What about this statement cannot be right and why?

(b) You state that a linear system can *only* have zero, one, or infinitely many solutions. Your friend, who hasn’t taken linear algebra, disagrees. They say that a parabola can intersect a line twice, meaning there are two solutions for that system and thus, making your statement false. Though his statement is true, why is his claim irrelevant to your statement?

- (c) Your friend just made \$500. He intends to invest in some stocks and purchase some snacks (they are \$ $\frac{5}{lb}$). Though you may think your friend made some weird choices, you set out to find how many combinations of items your friend can purchase. You create a linear system and find out that the system is (dependent / independent) and (consistent / inconsistent). [Circle the answers in parentheses]
- (d) On what basis did you make your choices for part (c)?

3. Assume that all of the dots (labelled A, B, C, D) must equal 0. Anything entering is positive and anything leaving is negative.



- (a) If one of your x_i turned out to be negative, what would this mean?

- (b) Write a system of equations for the diagram.

$$\left\{ \begin{array}{cccccc} + & + & + & = \\ + & + & + & = \\ + & + & + & = \\ + & + & + & = \end{array} \right.$$

- (c) Solve the system to find x_1, x_2, x_3, x_4 .

- (d) Write your answer from part (c) and write it in vector form.

$$\vec{x} = \begin{bmatrix} \\ \end{bmatrix}$$

4. **(Proof).** If $\mathbf{A}$ is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), prove that $\det(\mathbf{A})$ is the product of the entries on the main diagonal of the matrix; that is, $\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}$.

5. Given the following matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 6 & 10 \\ 6 & -5 & 7 \\ 10 & 4 & -3 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 4 & -2 & 1 \\ 2 & 3 & 6 \\ 5 & 7 & 1 \end{bmatrix},$$

answer the following problems.

- (a) Find $\det(\mathbf{A})$.

(b) Find $\det(\mathbf{A}^{-1})$.

(c) Find $\mathbf{A}^{-1}$.

6. Let T be a transformation defined by $T(x_1, x_2) = (x_1 + x_2, 2x_2, 3x_1 - 4x_2)$.

(a) Denote the transformation in the $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ notation.

(b) **(Proof)**. Prove that the transformation T follows the homogeneity property.

(c) **(Proof)**. Prove that the transformation T follows the additivity property.

- (d) Is the transformation T a ...
- i. linear transformation? _____
 - ii. matrix transformation? _____
- (e) Find the standard matrix $\mathbf{A}$ of the transformation T .
- (f) Find $T(6, 7)$.
- (g) If T were to be defined as $T(x_1, x_2) = (x_1^2, x_2^2, 3x_1 - 4x_2)$, why *wouldn't* this be a linear transformation? (I promise this is the last part)